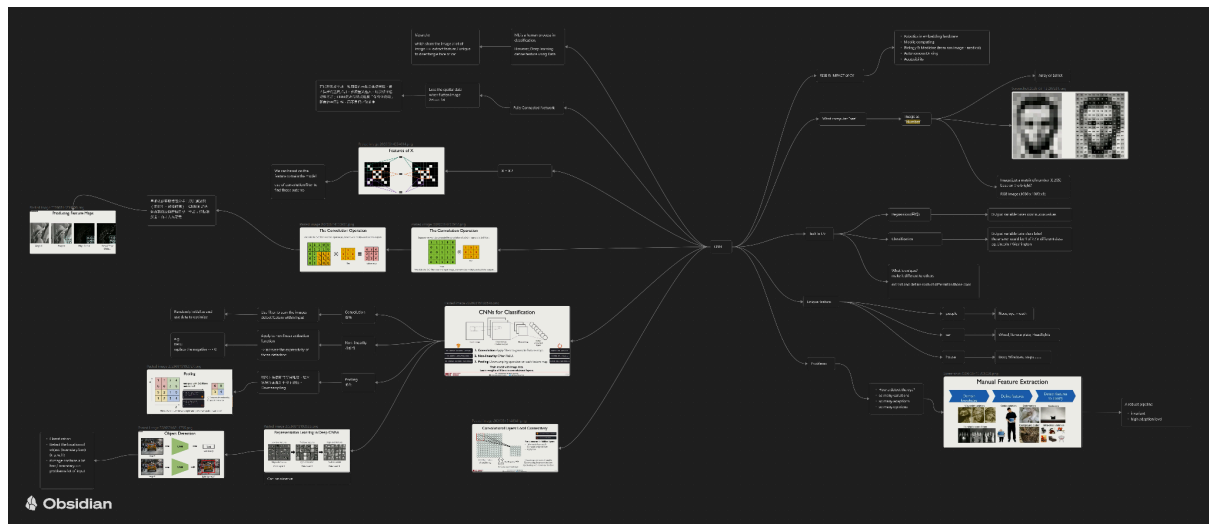


Reflection 3: Understanding of CNN

<https://www.youtube.com/watch?v=oGpzWAlP5p0>



As part of the project development I saw the lecture given by MIT called 6.S191; it was about Convolutional Neural Networks (CNNs) and it clarified for me why they are appropriate for use in computer vision. This was particularly useful to me when working on my deepfake image detection research as I identified how a CNN stores an image as an array of numbers and retains the spatial relationship between the pixels compared to a fully connected network.

The lecture included important concepts about what constitutes a CNN such as convolution, filters, ReLU, pooling, local connectivity and weight sharing. Additionally, it demonstrated how simple features are detected by CNNs in the first few layers and combined to form more complex features by the deeper layers which helped to provide a valid hypothesis which can explain why a ResNet50 based CNN would be able to find the visual artefacts present in fake images.

My thinking about my project changed as I recognised that the output from the CNN is not binary (real or fake) but it learns from the patterns of images present in the dataset. Thus, I believe that the quality of the images in the dataset and the way in which they have been pre-processed (cleaning and preparing) along with the way in which the CNN generalises to the 'real world' will all be very significant. Additionally, this activity helped me in better preparing for the explanations I will provide in both my report and viva, especially in terms

of being able to discuss why CNNs are suitable for detecting deepfake images and their limitations.